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MRQ3 applies the same shared evidence-intensity scale. The scope matrix codes how visibly each application scope structures the survey; the subsystem matrix codes how visibly each subsystem is covered; and the gap-signal matrix codes how visibly each subsystem is framed as under-reviewed or requiring future work.

Note on the normalised subsystem list. The backfill flagged in earlier drafts of this audit trail has been performed, and this note records what changed. Every one of the twenty-two surveys now reports its subsystem coverage on a single shared nine-subsystem list: HVAC; domestic hot water and water systems; lighting and daylighting; appliances and plug loads; electric vehicles including vehicle-to-home and vehicle-to-grid coupling; battery and electrical storage; photovoltaic generation and other renewables; indoor air quality and ventilation; and windows and blinds. Twenty-nine cells that previously had no column in their survey’s per-survey coverage table were coded from the source publications during revision rather than being carried forward as zeros; every one of those twenty-nine corrections is upward, and each is itemised with the supporting quotation in the revision log at the end of this file. Non-battery, thermal and phase-change-material storage has been dropped as a coded dimension, because it was coded in only four of the twenty-two surveys and the remaining eighteen values were never established. Whole-building or HEMS/BEMS scope, grid, demand-response and microgrid interaction, and multi-zone, district or cluster coordination are now reported in separate coordination tables alongside each coverage table, because these are levels of coordination rather than building subsystems and belong against their own denominator. The gap-signal tables are unchanged and deliberately retain their per-survey column lists.

1 Wang and Hong

Table 1: MRQ3 recoded audit table for Wang & Hong (2020). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	HVAC scope	Occupant scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
WangHong2020	3	2	0	3	2	The survey has a broad building-control scope and reviews RL applications across multiple building-control subjects. Broad scope receives score 3 because the review is explicitly framed around RL for building controls rather than a single subsystem. HVAC scope receives score 2 because HVAC is the most visible subject of control in the reviewed literature, but the survey itself is not limited to HVAC. Occupant-behavior scope receives score 0 because occupant behavior is not the declared scope of the paper. Multi-subsystem scope receives score 3 because the paper covers HVAC, batteries, home-appliance scheduling, domestic hot water, thermal energy storage, combined heat and power, windows, lighting, and related building-control domains. Deployment scope receives score 2 because real-building implementation is discussed, including the limited share of RL controllers implemented in real buildings, but deployment is not the sole organizing focus of the review.

Table 2: MRQ3 survey-scope scores for Wang & Hong (2020), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	HVAC-focused scope	Occupant-behavior scope	Multi-subsystem scope	Deployment/real-building scope
Wang & Hong (2020)	3	2	0	3	2

Table 3: MRQ3 subsystem-coverage scores for Wang & Hong (2020), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Wang & Hong (2020)	3	2	1	2	0	2	1	1	1

Table 4: MRQ3 coordination-level and external-interface coverage for Wang & Hong (2020), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Wang & Hong (2020)	2	1	1

Table 5: MRQ3 gap-signal scores for Wang & Hong (2020), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Windows
Wang & Hong (2020)	0	0	2	0	1	0	1	1	3	3	3	0

Table 6: MRQ3 summary table for Wang & Hong (2020), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Wang & Hong (2020)	Broad and HVAC-heavy	HVAC is the most visible subsystem, while batteries and home-appliance scheduling appear as secondary recurring domains. Non-thermal and larger-scale coordination areas are less consistently developed.	The paper supports a conservative MRQ3 reading in which HVAC dominance is a recurring survey-level pattern, while visual comfort, IAQ, occupant-feedback-based control, multi-agent community/campus/urban applications, and grid-operation applications remain comparatively under-reviewed or are mainly framed as future needs.

Table 7: One-line MRQ3 answer for Wang & Hong (2020), after recoding.

survey_id	MRQ3 answer
WangHong2020	Wang & Hong (2020) show that RL-for-building-controls survey coverage is broad but HVAC-heavy. Batteries and appliance scheduling appear as secondary recurring domains, while visual comfort, IAQ, occupant-feedback-based control, and multi-agent community-, campus-, urban-, and grid-oriented applications are less consistently covered and are often framed as future needs.

2 Han et al.

Table 8: MRQ3 recoded audit table for Han et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	HVAC scope	Occupant scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
HanEtAl2021	0	2	3	1	0	The survey is specialized around occupant behavior in building control rather than broad building-control coverage. Broad scope receives score 0 because the review does not aim to cover all RL-for-building subsystems. HVAC scope receives score 2 because HVAC dominates the occupant-behavior RL studies reviewed, although the paper is not declared as HVAC-only. Occupant-behavior scope receives score 3 because occupant behavior is the declared and structuring scope of the review. Multi-subsystem scope receives score 1 because lighting, windows, vessel or water-heating-related control, and one photovoltaic-related case appear, but these remain minor compared with HVAC. Deployment scope receives score 0 because real-building deployment is not a central scope dimension of the review.

Table 9: MRQ3 survey-scope scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	HVAC-focused scope	Occupant-behavior scope	Multi-subsystem scope	Deployment/real-building scope
Han et al. (2021)	0	2	3	1	0

Table 10: MRQ3 subsystem-coverage scores for Han et al. (2021), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Han et al. (2021)	3	1	1	1	0	0	1	2	1

Table 11: MRQ3 coordination-level and external-interface coverage for Han et al. (2021), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Han et al. (2021)	–	–	–

Table 12: MRQ3 gap-signal scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	PV/renewables	Windows
Han et al. (2021)	0	3	3	0	3

Table 13: MRQ3 summary table for Han et al. (2021), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Han et al. (2021)	Specialized and HVAC-heavy	Within the occupant-behavior RL subliterature, HVAC is the dominant subsystem. Lighting, window, and vessel or water-heating-related controls appear, but are explicitly characterized as relatively uncommon.	The paper supports an MRQ3 reading in which occupant-behavior-aware RL remains concentrated on HVAC, while non-HVAC applications are secondary and are framed more as future research directions than as mature, well-covered review areas.

Table 14: One-line MRQ3 answer for Han et al. (2021), after recoding.

survey_id	MRQ3 answer
HanEtAl2021	Han et al. (2021) show a clear HVAC-dominant pattern within RL studies on occupant behavior in building control. Lighting, windows, and vessel or water-heating-related applications are present but minor, and the paper explicitly frames these non-HVAC directions as relatively uncommon areas requiring further study.

3 Shaqour and Hagishima

Table 15: MRQ3 recoded audit table for Shaqour & Hagishima (2022). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	HVAC scope	Building-type scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
ShaqourHagishima2022	3	2	3	3	2	The survey has a broad DRL-based BEMS scope and is structured around multiple building types. Broad scope receives score 3 because the paper systematically reviews DRL-based energy management across building applications rather than a single subsystem. HVAC scope receives score 2 because HVAC is highly recurring across building types, but the paper is not limited to HVAC. Building-type scope receives score 3 because residential, office, educational, data-centre, and other building categories structure the review. Multi-subsystem scope receives score 3 because the paper compares controlled appliances and systems, renewable-energy integration, demand response, PV, ESS, EVs, appliances, TES, blinds, lighting, and related building-control elements across building types. Deployment scope receives score 2 because real-world implementation and validation are discussed, including the limited share of real implementations, but deployment is not the sole organizing scope of the review.

Table 16: MRQ3 survey-scope scores for Shaqour & Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	HVAC-focused scope	Building-type/MORL scope	Multi-subsystem scope	Deployment/real-building scope
Shaqour & Hagishima (2022)	3	2	3	3	2

Table 17: MRQ3 subsystem-coverage scores for Shaqour & Hagishima (2022), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Shaqour & Hagishima (2022)	3	1	1	2	1	2	2	1	1

Table 18: MRQ3 coordination-level and external-interface coverage for Shaqour & Hagishima (2022), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Shaqour & Hagishima (2022)	3	2	2

Table 19: MRQ3 gap-signal scores for Shaqour & Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Other niche/Windows
Shaqour & Hagishima (2022)	0	1	1	0	2	2	2	0	2	1	3	1

Table 20: MRQ3 summary table for Shaqour & Hagishima (2022), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Shaqour & Hagishima (2022)	Broad, building-type-centered, and HVAC-heavy	HVAC and cooling are the most consistently visible control targets across building types. Residential studies show the broadest non-HVAC spread through demand response, ESS, PV, appliances, and some EV integration, while office, educational, and especially data-centre studies are more thermally centered.	The paper supports an MRQ3 reading in which HVAC remains the most recurrent subsystem across DRL-BEMS surveys, but subsystem diversity depends strongly on building type. Residential studies diversify into DR, PV, ESS, and appliances, whereas EVs, renewable/storage/DR integration in data centres, and multi-building or district-scale coordination remain weaker or explicitly less explored.

Table 21: One-line MRQ3 answer for Shaqour & Hagishima (2022), after recoding.

survey_id	MRQ3 answer
ShaqourHagishima2022	Shaqour & Hagishima (2022) show that DRL-BEMS coverage is broad but still HVAC-heavy across building types. Residential studies have the widest subsystem spread, including demand response, PV, ESS, appliances, and some EV integration, while office, educational, and data-centre studies remain more thermally centered and multi-building or district-scale coordination is explicitly less explored.

4 Al Sayed et al.

Table 22: MRQ3 recoded audit table for Al Sayed et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	HVAC scope	Building-type/MORL scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
AlSayedEtAl2024	0	3	0	1	2	The survey is explicitly specialized around reinforcement learning for HVAC control rather than broad subsystem comparison. Broad scope receives score 0 because the review does not aim to cover all RL-for-building subsystems. HVAC scope receives score 3 because HVAC is the declared and structuring scope of the review. Building-type/MORL scope receives score 0 because building type and MORL are not the paper’s main organizing axes. Multi-subsystem scope receives score 1 because air quality, CO ₂ , domestic hot water, demand response, PV, and battery-related elements appear only as auxiliary HVAC-linked cases rather than coequal subsystem domains. Deployment scope receives score 2 because real-building implementation, simulation-to-reality mismatch, and generalization are discussed, but deployment is not the declared scope of the survey.

Table 23: MRQ3 survey-scope scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	HVAC-focused scope	Building-type/MORL scope	Multi-subsystem scope	Deployment/real-building scope
Al Sayed et al. (2024)	0	3	0	1	2

Table 24: MRQ3 subsystem-coverage scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Al Sayed et al. (2024)	3	1	0	0	0	1	1	1	0

Table 25: MRQ3 coordination-level and external-interface coverage for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Al Sayed et al. (2024)	0	1	0

Table 26: MRQ3 gap-signal scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Other niche/Windows
Al Sayed et al. (2024)	0	1	0	0	0	1	1	0	1	1	0	0

Table 27: MRQ3 summary table for Al Sayed et al. (2024), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Al Sayed et al. (2024)	Specialized and HVAC-centered	The review is HVAC-dominant by design. Non-HVAC elements such as air quality, CO ₂ , domestic hot water, demand response, PV, and battery-related cases appear mainly as auxiliary HVAC-linked formulation elements rather than as standalone subsystem domains.	For MRQ3, this paper should be interpreted as evidence of scope-driven HVAC concentration rather than as a broad subsystem survey. Its strongest explicit gaps concern real deployment and generalization, while subsystem-level omissions beyond HVAC are mostly outside the declared scope.

Table 28: One-line MRQ3 answer for Al Sayed et al. (2024), after recoding.

survey_id	MRQ3 answer
AlSayedEtAl2024	Al Sayed et al. (2024) is a specialized RL-for-HVAC review. It shows HVAC as a structuring scope, while IAQ, CO ₂ , domestic hot water, demand response, PV, and battery-related cases appear only as secondary HVAC-linked elements. Its main gaps concern real-building deployment and generalization rather than broad subsystem omission.

5 Huynh et al.

Table 29: MRQ3 recoded audit table for Huynh et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	HVAC scope	MORL scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
HuynhEtAl2025	3	2	3	3	2	The survey has a broad smart-building MORL scope. Broad scope receives score 3 because the paper systematically reviews MORL applications across smart-building domains. HVAC scope receives score 2 because HVAC is highly visible and recurrent, but it is not the sole declared scope. MORL scope receives score 3 because multi-objective reinforcement learning is the declared and structuring focus of the survey. Multi-subsystem scope receives score 3 because the paper’s application taxonomy includes HVAC, lighting, appliance scheduling, EMS, and other applications. Deployment scope receives score 2 because evaluation environments, virtual versus real-world validation, and field implementation are discussed, but deployment is not the sole organizing scope.

Table 30: MRQ3 survey-scope scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	HVAC-focused scope	Building-type/MORL scope	Multi-subsystem scope	Deployment/real-building scope
Huynh et al. (2025)	3	2	3	3	2

Table 31: MRQ3 subsystem-coverage scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Huynh et al. (2025)	3	0	1	2	0	1	1	1	1

Table 32: MRQ3 coordination-level and external-interface coverage for Huynh et al. (2025), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Huynh et al. (2025)	2	1	0

Table 33: MRQ3 gap-signal scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Other niche/Windows
Huynh et al. (2025)	0	2	2	0	2	0	0	0	0	2	0	3

Table 34: MRQ3 summary table for Huynh et al. (2025), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Huynh et al. (2025)	Broad MORL scope with HVAC concentration	The review is broad across smart-building MORL applications, but its most visible application cluster remains HVAC and thermal-comfort-oriented control. Appliance scheduling and EMS form the next substantial tier, while lighting is lighter and many niche smart-building applications remain peripheral.	The paper supports an MRQ3 reading in which MORL research remains centered on core energy-comfort control problems, especially HVAC, while broader application domains such as water-related objectives, visual/acoustic comfort, EV-specific concerns, and non-energy building-management functions are weakly represented or framed mainly as future opportunities.

Table 35: One-line MRQ3 answer for Huynh et al. (2025), after recoding.

survey_id	MRQ3 answer
HuynhEtAl2025	Huynh et al. (2025) portray MORL research in smart buildings as broad but still centered on HVAC and thermal-comfort-oriented control. EMS and appliance scheduling form the next substantive application areas, lighting remains lighter, and many other smart-building applications are peripheral or mainly proposed as future opportunities.

6 Zhang et al.

Table 36: MRQ3 recoded audit table for Zhang et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
ZhangEtAl2022	3	3	2	3	2	The survey has a broad building energy management scope and is structured around reinforcement learning, model predictive control, and hybrid BEMS control strategies. Broad scope receives score 3 because the paper frames BEMS as involving HVAC, lighting, household or commercial appliances, EVs, battery energy storage, renewable sources, and the power grid. Main specialized scope receives score 3 because BEMS control with RL, MPC, and RL-MPC combinations is the organizing focus of the survey. Secondary focus scope receives score 2 because HVAC and thermal control are recurring application strands, but the paper is not HVAC-only. Multi-subsystem scope receives score 3 because the paper discusses home energy management, appliance scheduling, domestic hot water, HVAC control, ESS and EV scheduling, commercial-building subsystem control, and multi-agent or community-scale energy management. Deployment scope receives score 2 because real-world exploration cost, simulator and model dependence, historical building data, and practical implementation difficulty are discussed, but deployment is not the sole organizing scope.

Table 37: MRQ3 survey-scope scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Zhang et al. (2022)	3	3	2	3	2

Table 38: MRQ3 subsystem-coverage scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Zhang et al. (2022)	2	1	1	2	2	2	1	1	0

Table 39: MRQ3 coordination-level and external-interface coverage for Zhang et al. (2022), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Zhang et al. (2022)	3	2	2

Table 40: MRQ3 gap-signal scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Other RES	Non-battery storage	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Windows
Zhang et al. (2022)	0	1	1	0	0	0	1	0	0	0	0	1	1	0

Table 41: MRQ3 summary table for Zhang et al. (2022), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Zhang et al. (2022)	Broad BEMS and method-centered	The survey covers a broad BEMS application space, with whole-home/building energy management, HVAC, appliances, EVs, ESS, grid/DR, and community-scale settings all visible. HVAC and integrated BEMS are the most recurring strands, but the paper does not make a strong subsystem-imbalance claim.	For MRQ3, this paper supports a cautious reading: RL and MPC research in BEMS is broader than HVAC alone, but still strongly organized around thermal control and integrated energy-management problems. Its clearest gaps are methodological—data efficiency, safety, generalization, simulator/model dependence, and learning from historical data—rather than subsystem-specific omissions.

Table 42: One-line MRQ3 answer for Zhang et al. (2022), after recoding.

survey_id	MRQ3 answer
ZhangEtAl2022	Zhang et al. (2022) presents a broad RL–MPC BEMS survey in which integrated home/building energy management and HVAC are the most visible application strands, alongside meaningful coverage of appliances, domestic hot water, EVs, ESS, renewables, grid/DR, and community-scale settings. Its clearest limitations are methodological rather than strong subsystem omissions.

7 Fu et al.

Table 43: MRQ3 recoded audit table for Fu et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
FuEtAl2022	3	3	2	2	2	The survey has a broad intelligent-building energy-efficiency scope and reviews RL applications across multiple building energy-consumption control objects. Broad scope receives score 3 because the paper frames building energy use through HVAC systems, water heating equipment, lifts, lighting systems, smart-home decision making, photovoltaic and smart-grid control, and other building control applications. Main specialized scope receives score 3 because RL for building energy-efficiency control is the declared organizing focus of the review. Secondary focus scope receives score 2 because HVAC is repeatedly presented as the main energy-consumption target, but the paper is not HVAC-only. Multi-subsystem scope receives score 2 because non-HVAC applications are discussed, including domestic water heaters, dynamic pricing and demand response, PV and smart-grid control, window-related control, smart-home decision making, battery/solar applications, and occupant-related control, but these are not organized as a full subsystem taxonomy. Deployment scope receives score 2 because real-building cost, training burden, unsafe exploration, and simulation-to-reality limitations are discussed, but deployment is not the declared organizing scope.

Table 44: MRQ3 survey-scope scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Fu et al. (2022)	3	3	2	2	2

Table 45: MRQ3 subsystem-coverage scores for Fu et al. (2022), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Fu et al. (2022)	3	2	1	1	0	1	1	1	1

Table 46: MRQ3 coordination-level and external-interface coverage for Fu et al. (2022), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Fu et al. (2022)	2	2	2

Table 47: MRQ3 gap-signal scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Other RES	Non-battery storage	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Windows
Fu et al. (2022)	0	1	1	1	0	1	1	0	0	0	1	2	3	1

Table 48: MRQ3 summary table for Fu et al. (2022), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Fu et al. (2022)	Broad but HVAC-centered	HVAC is repeatedly presented as the main energy-consumption and RL-control target, while water heating, lighting/window-related control, smart-home energy management, PV, smart-grid/DR, battery/solar, and occupant-related applications appear as secondary or scattered domains.	The paper supports an MRQ3 reading in which HVAC remains the dominant reviewed subsystem within building energy-efficiency control, but non-HVAC applications are present. Its clearest gap signals concern multi-agent control for complex multi-object or larger-scale systems and occupant-related control, rather than sharply defined omission of one subsystem.

Table 49: One-line MRQ3 answer for Fu et al. (2022), after recoding.

survey_id	MRQ3 answer
FuEtAl2022	Fu et al. (2022) reviews a broad range of RL applications for building energy-efficiency control, but HVAC remains the most visible subsystem. Water heating, lighting/window-related control, smart-home and grid-oriented management, PV, battery/solar, and occupant-related applications appear as secondary domains, while larger-scale multi-agent control is framed as a stronger future need.

8 Michailidis et al.

Table 50: MRQ3 recoded audit table for Michailidis et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
MichailidisEtAl2025	2	3	3	3	2	The survey is specialized around RL for RES-integrated BEMS, while still covering multiple building-energy subsystems within that scope. Broad scope receives score 2 because the paper is broad within RES-integrated BEMS but is not a generic all-subsystem RL-for-buildings survey. Main specialized scope receives score 3 because RES-integrated BEMS is the declared and structuring scope. Secondary focus scope receives score 3 because PV/solar integration is the dominant RES pattern and structures much of the synthesis. Multi-subsystem scope receives score 3 because the review includes PV, wind turbines, solar water heaters, ground heat pumps, biomass systems, HVAC, domestic hot water, lighting, ESS, EVs, whole-building BEMS, and grid-related applications. Deployment scope receives score 2 because limited real-life experiments and simulation-heavy validation are discussed, but deployment is not the sole organizing scope.

Table 51: MRQ3 survey-scope scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Michailidis et al. (2025)	2	3	3	3	2

Table 52: MRQ3 subsystem-coverage scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Michailidis et al. (2025)	2	1	1	2	1	2	3	1	1

Table 53: MRQ3 coordination-level and external-interface coverage for Michailidis et al. (2025), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Michailidis et al. (2025)	3	2	1

Table 54: MRQ3 gap-signal scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	EV/V2H/V2G	Battery/ESS	PV/renewables	Other RES	Non-battery storage	Whole-home/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Michailidis et al. (2025)	0	1	1	1	0	0	2	3	0	1	3

Table 55: MRQ3 summary table for Michailidis et al. (2025), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Michailidis et al. (2025)	RES/BEMS-specialized with PV concentration	The review is centered on RES-integrated BEMS, with PV/solar and whole-building BEMS as structuring domains and battery storage as a major recurring component. HVAC appears as a substantial coupled subsystem, while other RES technologies and non-battery storage forms are less visible.	The paper supports an MRQ3 reading in which RL research on renewable-energy utilization in buildings is concentrated on PV-driven, battery-aware, whole-building energy management, with HVAC as a recurring companion subsystem. Other RES technologies, non-battery storage forms, and larger-scale district or microgrid coordination remain weaker or explicitly framed as future priorities.

Table 56: One-line MRQ3 answer for Michailidis et al. (2025), after recoding.

survey_id	MRQ3 answer
MichailidisEtAl2025	Michailidis et al. (2025) portray RL research on renewable-energy utilization in buildings as concentrated on RES-integrated BEMS, especially PV/solar and battery-coupled configurations. HVAC is a substantial coupled subsystem, while other RES technologies, non-battery storage, and larger-scale district or microgrid coordination remain less developed or are explicitly identified as future needs.

9 Yu et al.

Table 57: MRQ3 recoded audit table for Yu et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
YuEtAl2021	3	3	2	3	2	The survey has a broad smart-building energy-management scope and is explicitly organized around DRL for SBEM. Broad scope receives score 3 because the paper reviews smart building energy-management applications rather than a single subsystem. Main specialized scope receives score 3 because DRL-based SBEM is the declared and structuring focus of the paper, organized around single building energy subsystems, multiple energy subsystems, and building microgrids. Secondary focus scope receives score 2 because HVAC receives strong attention at the single-subsystem level, but the survey is not HVAC-only. Multi-subsystem scope receives score 3 because the paper covers PV, wind turbines, distributed generation, electric and thermal energy storage, HVAC, lighting, blind and window systems, electric water heaters, EVs, appliances, and building microgrids. Deployment scope receives score 2 because simulation-heavy validation and practical implementation limitations are discussed, but deployment is not the sole organizing scope.

Table 58: MRQ3 survey-scope scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Yu et al. (2021)	3	3	2	3	2

Table 59: MRQ3 subsystem-coverage scores for Yu et al. (2021), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Yu et al. (2021)	3	2	1	2	2	2	2	1	1

Table 60: MRQ3 coordination-level and external-interface coverage for Yu et al. (2021), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Yu et al. (2021)	3	2	2

Table 61: MRQ3 gap-signal scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	HVAC	Temperature/thermal comfort	Humidity	IAQ/ventilation	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster	Windows	Thermal storage	Building clusters/local energy communities	Multi-agent BES control
Yu et al. (2021)	0	0	0	1	0	1	0	0	0	0	0	0	2	1	0	0	0

Table 62: MRQ3 summary table for Yu et al. (2021), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Yu et al. (2021)	Broad SBEM and scale-based	The review is organized by system scale rather than by a single subsystem. HVAC is the clearest recurring strand at the single-subsystem level, while integrated SBEM and microgrid-oriented settings broaden coverage toward PV, ESS, EVs, appliances, lighting, electric water heaters, and grid interaction.	For MRQ3, the paper supports a balanced reading: DRL-based SBEM research is HVAC-heavy at the single-subsystem level, but broader at whole-building and microgrid scales. The strongest gaps are more methodological and scalability-oriented than sharply subsystem-specific.

Table 63: One-line MRQ3 answer for Yu et al. (2021), after recoding.

survey_id	MRQ3 answer
YuEtAl2021	Yu et al. (2021) review DRL for SBEM through a scale-based lens and show that HVAC dominates the single-subsystem literature, while broader SBEM work substantially covers integrated PV, ESS, EV, appliance, lighting, hot-water, and microgrid configurations. The survey is therefore HVAC-heavy but clearly broader than a dedicated HVAC-only review.

10 Sierla et al.

Table 64: MRQ3 recoded audit table for Sierla et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
SierlaEtAl2022	0	3	3	1	2	The survey is specialized around RL applications to HVAC control rather than broad subsystem coverage. Broad scope receives score 0 because the paper does not aim to review all RL-for-building subsystems. Main specialized scope receives score 3 because HVAC control is the declared and structuring scope. Secondary focus scope receives score 3 because the paper categorizes the literature by zone applicability, action formulation, and indoor-environment variable being controlled, including temperature, humidity, and air quality. Multi-subsystem scope receives score 1 because the paper is not multi-subsystem in the usual building-device sense; its breadth is mainly within HVAC variables and zone structures. Deployment scope receives score 2 because deployability, action abstraction, real HVAC interfacing, and scalability are discussed, but deployment is not the sole organizing scope.

Table 65: MRQ3 survey-scope scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Sierla et al. (2022)	0	3	3	1	2

Table 66: MRQ3 subsystem-coverage scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Sierla et al. (2022)	3	2	1	0	0	1	2	1	2

Table 67: MRQ3 coordination-level and external-interface coverage for Sierla et al. (2022), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Sierla et al. (2022)	–	–	–

Table 68: MRQ3 gap-signal scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	HVAC	Temperature/thermal comfort	Humidity	IAQ/ventilation	Single-zone HVAC	Independent-zone HVAC	Interdependent multi-zone HVAC
Sierla et al. (2022)	0	0	1	3	0	1	2

Table 69: MRQ3 summary table for Sierla et al. (2022), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Sierla et al. (2022)	Specialized HVAC with internal imbalance	The review shows a clear concentration on temperature-based thermal-comfort control within RL-for-HVAC research. Humidity and especially air-quality-oriented formulations are less visible, while interdependent multi-zone control remains challenging because of scalability and realistic HVAC interfacing.	For MRQ3, the paper should be interpreted as evidence that HVAC-oriented RL surveys can still be internally imbalanced: temperature-based formulations dominate, while broader indoor-environment variables such as IAQ and humidity receive weaker attention. Its omissions are therefore within HVAC scope, not across unrelated building subsystems.

Table 70: One-line MRQ3 answer for Sierla et al. (2022), after recoding.

survey_id	MRQ3 answer
SierlaEtAl2022	Sierla et al. (2022) show that within RL-for-HVAC research, temperature-based thermal-comfort control dominates the reviewed literature, while humidity and especially air-quality-oriented HVAC control are less developed. The paper’s MRQ3 value is therefore mainly in exposing internal imbalance within HVAC-focused RL reviews.

11 Latoń et al.

Table 71: MRQ3 recoded audit table for Latoń et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
LatonEtAl2024	0	3	3	3	2	The survey is specialized around RL and DRL for Home Energy Management Systems rather than broad all-building subsystem coverage. Broad scope receives score 0 because the paper is HEMS-specific. Main specialized scope receives score 3 because HEMS is the declared and structuring scope. Secondary focus scope receives score 3 because demand response, load scheduling, renewable-energy integration, IoT-enabled automation, and residential/prosumer microgrids structure the paper’s application framing. Multi-subsystem scope receives score 3 because the paper discusses HVAC systems, lighting, appliances, electric vehicles, solar panels, battery storage, and smart-grid interaction as HEMS-managed components. Deployment scope receives score 2 because scalability, privacy, data security, heterogeneous system integration, and the need for real applications and case studies are discussed, but deployment is not the sole organizing scope.

Table 72: MRQ3 survey-scope scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Latoń et al. (2024)	0	3	3	3	2

Table 73: MRQ3 subsystem-coverage scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Latoń et al. (2024)	1	0	1	2	1	2	2	1	0

Table 74: MRQ3 coordination-level and external-interface coverage for Latoń et al. (2024), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Latoń et al. (2024)	3	2	–

Table 75: MRQ3 gap-signal scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	HVAC	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid
Latoń et al. (2024)	1	1	0	1	0	0	0	0

Table 76: MRQ3 summary table for Latoń et al. (2024), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Latoń et al. (2024)	Specialized HEMS orchestration	The review is centered on whole-home energy-management functions, especially demand response, load scheduling, renewable integration, energy storage, smart-grid interaction, and residential/prosumer microgrids. Individual subsystems such as HVAC and lighting appear mainly as components inside broader HEMS coordination.	For MRQ3, the paper shows that not all recent RL-for-buildings surveys are HVAC-heavy. Some are instead centered on HEMS-level orchestration, where the main application bias shifts toward scheduling, DSM/DSR, RES-storage coordination, and smart-grid integration. The main gaps are deployment and infrastructure related rather than subsystem-specific.

Table 77: One-line MRQ3 answer for Latoń et al. (2024), after recoding.

survey_id	MRQ3 answer
LatonEtAl2024	Latoń et al. (2024) portray RL/DRL research in HEMS as centered on whole-home orchestration, especially demand response, load scheduling, renewable integration, storage management, and smart-grid interaction. HVAC, lighting, and other individual subsystems appear mainly as components within that broader HEMS frame rather than as standalone review targets.

12 Weinberg et al.

Table 78: MRQ3 recoded audit table for Weinberg et al. (2023). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
WeinbergEtAl2023	0	3	3	1	2	The survey is specialized around RL for Building Energy Systems from a computer-science perspective, with BES narrowed mainly toward thermal systems. Broad scope receives score 0 because this is not a broad all-subsystem review. Main specialized scope receives score 3 because thermal BES and HVAC-related control define the paper’s application scope. Secondary focus scope receives score 3 because the survey is explicitly structured around RL in single buildings, RL in building clusters, and multi-agent aspects. Multi-subsystem scope receives score 1 because the breadth is mainly scale-based rather than device-category-based; diverse non-thermal subsystems such as lighting, appliances, and EVs are not a major part of the review. Deployment scope receives score 2 because transfer learning, sim-to-real mismatch, differing state–action spaces across buildings, data limitations, and practical implementation challenges are discussed, but deployment is not the only organizing scope.

Table 79: MRQ3 survey-scope scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Weinberg et al. (2023)	0	3	3	1	2

Table 80: MRQ3 subsystem-coverage scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Weinberg et al. (2023)	3	1	1	1	0	0	2	2	1

Table 81: MRQ3 coordination-level and external-interface coverage for Weinberg et al. (2023), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Weinberg et al. (2023)	2	–	2

Table 82: MRQ3 gap-signal scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	HVAC	Thermal storage	DHW/water	Whole-building thermal BES	Building clusters/local energy communities	Multi-agent BES control
Weinberg et al. (2023)	0	0	1	0	3	3

Table 83: MRQ3 summary table for Weinberg et al. (2023), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Weinberg et al. (2023)	Specialized thermal-BES and scale-based	The review is concentrated on thermal BES and HVAC-related control, especially heating, cooling, ventilation, and thermal storage. Its main broadening beyond single-building thermal control occurs through building-cluster control and multi-agent coordination rather than through extensive coverage of diverse non-thermal subsystems.	For MRQ3, the paper shows that some RL-for-buildings reviews remain strongly centered on thermal system control, with the main next frontier being larger-scale coordination across buildings rather than balanced coverage of lighting, appliances, EVs, and other non-thermal device categories.

Table 84: One-line MRQ3 answer for Weinberg et al. (2023), after recoding.

survey_id	MRQ3 answer
WeinbergEtAl2023	Weinberg et al. (2023) present a thermal-BES- and HVAC-centered RL review that extends mainly through building-cluster and multi-agent perspectives. The paper therefore shows stronger survey attention to heating, cooling, ventilation, and thermal storage than to a broad range of non-thermal building subsystems.

13 Al-Ani and Das

Table 85: MRQ3 recoded audit table for Al-Ani & Das (2022). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
AlAniDas2022	0	3	2	3	2	The survey is specialized around reinforcement learning for home energy management systems rather than broad all-building subsystem coverage. Broad scope receives score 0 because the paper is HEMS-specific. Main specialized scope receives score 3 because HEMS is the declared and structuring scope. Secondary focus scope receives score 2 because HVAC, fans, and water heaters form the largest single application class, but the survey is not limited to this class. Multi-subsystem scope receives score 3 because the paper divides the reviewed literature into multiple application classes, including HVAC/fans/water heaters, EVs, energy storage, renewable generation, other loads, demand response, and peer-to-peer trading. Deployment scope receives score 2 because scalability, data security, and the need for more real applications and case studies are discussed, but deployment is not the only organizing scope.

Table 86: MRQ3 survey-scope scores for Al-Ani & Das (2022), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Al-Ani & Das (2022)	0	3	2	3	2

Table 87: MRQ3 subsystem-coverage scores for Al-Ani & Das (2022), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Al-Ani & Das (2022)	3	3	0	2	2	2	2	0	0

Table 88: MRQ3 coordination-level and external-interface coverage for Al-Ani & Das (2022), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Al-Ani & Das (2022)	3	2	0

Table 89: MRQ3 gap-signal scores for Al-Ani & Das (2022), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	P2P trading
Al-Ani & Das (2022)	0	0	0	0	1	1	1	0	0	0	0	2

Table 90: MRQ3 summary table for Al-Ani & Das (2022), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Al-Ani & Das (2022)	Specialized HEMS with multi-class coverage	The review is centered on HEMS orchestration, but within that frame the largest single application class is HVAC, fans, and water heaters. EV/storage/renewable generation and demand response are also substantial recurring domains, while peer-to-peer trading is a smaller tail of the reviewed literature.	For MRQ3, the paper shows that RL-for-HEMS surveys are not purely HVAC-only, but still give the strongest single-class emphasis to thermal-device control. They broaden toward storage, renewables, demand response, and grid-facing coordination, while peer-to-peer trading remains less visibly developed.

Table 91: One-line MRQ3 answer for Al-Ani & Das (2022), after recoding.

survey_id	MRQ3 answer
AlAniDas2022	Al-Ani & Das (2022) portray RL research in HEMS as centered on whole-home orchestration, with HVAC/fans/water-heater control as the largest single application class, substantial additional coverage of EV/storage/renewable generation and demand response, and comparatively limited attention to peer-to-peer energy trading.

14 Mason and Grijalva

Table 92: MRQ3 recoded audit table for Mason & Grijalva (2019). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
MasonGrijalva2019	3	3	2	3	2	The survey has a broad autonomous building energy management scope and reviews reinforcement learning applications across several building-energy domains. Broad scope receives score 3 because the paper is framed as a comprehensive review of RL for autonomous building energy management. Main specialized scope receives score 3 because autonomous building energy management is the organizing focus of the review. Secondary focus scope receives score 2 because HVAC is the strongest and most recurrent application strand, but the paper is broader than HVAC-only. Multi-subsystem scope receives score 3 because the paper covers HVAC, water heaters, home management systems, lighting, appliances, PV, batteries, EVs, and smart-grid interaction. Deployment scope receives score 2 because simulation dependence, simulator design, and real-world robustness are discussed, but deployment is not the sole organizing scope.

Table 93: MRQ3 survey-scope scores for Mason & Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Mason & Grijalva (2019)	3	3	2	3	2

Table 94: MRQ3 subsystem-coverage scores for Mason & Grijalva (2019), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Mason & Grijalva (2019)	3	2	1	2	1	1	1	1	1

Table 95: MRQ3 coordination-level and external-interface coverage for Mason & Grijalva (2019), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Mason & Grijalva (2019)	2	1	1

Table 96: MRQ3 gap-signal scores for Mason & Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	P2P trading
Mason & Grijalva (2019)	0	0	1	0	1	1	1	0	0	1	3	0

Table 97: MRQ3 summary table for Mason & Grijalva (2019), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Mason & Grijalva (2019)	Broad early BEM and HVAC-led	The review is clearly HVAC-led, with water heaters as a meaningful secondary domain and broader home management systems extending coverage to appliances, lighting, PV, batteries, EVs, and smart-grid interaction. The paper does not provide a strong quantitative subsystem-bias analysis, but its narrative emphasis is visibly heavier on HVAC than on other domains.	For MRQ3, the paper supports a reading in which early RL-for-building-energy-management surveys were dominated by HVAC control, while water heating and multi-device home management formed secondary strands. The clearest future need is scaling toward communities of homes rather than balancing coverage across every subsystem.

Table 98: One-line MRQ3 answer for Mason & Grijalva (2019), after recoding.

survey_id	MRQ3 answer
MasonGrijalva2019	Mason & Grijalva (2019) present a broad RL-for-building-energy-management review in which HVAC is the dominant application area, water heaters form a clear secondary domain, and broader home-management studies extend coverage to appliances, lighting, PV, batteries, and EV-related coordination without displacing HVAC as the main emphasis.

15 Pinthurat et al.

Table 99: MRQ3 recoded audit table for Pinthurat et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
PinthuratEtAl2024	0	3	3	3	3	The survey is specialized around reinforcement-learning-based smart home energy management systems with energy storage. Broad scope receives score 0 because the paper is not a broad all-building subsystem survey. Main specialized scope receives score 3 because SHEMS with energy storage is the declared and structuring scope. Secondary focus scope receives score 3 because energy-storage integration structures the entire review, including battery type, state of charge, state of health, life cycle, degradation, storage constraints, cost, efficiency, operation mode, and pricing mechanisms. Multi-subsystem scope receives score 3 because the smart-home architecture includes energy storage, renewable energy, plug-in EVs, controllable appliances such as AC and water heaters, uncontrollable appliances, smart meters, and grid interaction. Deployment scope receives score 3 because real-world testing and validation are treated as a dedicated future direction, with field trials emphasized as essential for bridging theory and practice.

Table 100: MRQ3 survey-scope scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Pinthurat et al. (2024)	0	3	3	3	3

Table 101: MRQ3 subsystem-coverage scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Pinthurat et al. (2024)	1	1	1	2	1	3	2	1	0

Table 102: MRQ3 coordination-level and external-interface coverage for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Pinthurat et al. (2024)	3	2	1

Table 103: MRQ3 gap-signal scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster	P2P trading
Pinthurat et al. (2024)	1	1	0	1	0	0	0	3	3	0

Table 104: MRQ3 summary table for Pinthurat et al. (2024), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Pinthurat et al. (2024)	Specialized SHEMA and storage-centered	The review is centered on energy-storage-integrated smart-home orchestration, with battery/energy storage as the most visible application component and strong linkage to renewable integration, demand-side management, and grid-aware operation. HVAC, AC, and water heaters appear mainly as controllable appliances inside the smart-home architecture rather than as standalone review domains.	For MRQ3, the paper shows that some recent RL-for-homes reviews are not HVAC-dominant, but instead storage-centered. The clearest under-reviewed directions are demand-side management integration and multi-agent or multi-home coordination, both framed as future research needs.

Table 105: One-line MRQ3 answer for Pinthurat et al. (2024), after recoding.

survey_id	MRQ3 answer
PinthuratEtAl2024	Pinthurat et al. (2024) portray RL research in smart homes as centered on energy-storage-integrated SHEMA orchestration, with battery/energy storage as the structuring application focus, strong coupling to renewable integration and grid-aware management, and comparatively limited use of multi-agent or multi-home coordination approaches.

16 Kannari et al.

Table 106: MRQ3 recoded audit table for Kannari et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
KannariEtAl2025	0	3	3	1	3	The survey is specialized around reinforcement-learning-based control and optimization in real buildings rather than broad all-building subsystem coverage. Broad scope receives score 0 because the paper is not a broad subsystem-by-subsystem review. Main specialized scope receives score 3 because real-building RL deployment is the declared and structuring scope of the review. Secondary focus scope receives score 3 because the reviewed real-building deployment literature is strongly organized around HVAC, heating, cooling, ventilation, and related thermal services. Multi-subsystem scope receives score 1 because the paper includes only scattered broader cases, such as domestic hot water, windows plus air cleaning, HVAC plus lighting and plug loads, and one floor-heating case coupled with an EV. Deployment scope receives score 3 because deployment in actual buildings is central to the paper, including real-world implementation conditions, control interval, deployment mode, validation setting, pretraining, safety, and practical adoption concerns.

Table 107: MRQ3 survey-scope scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Kannari et al. (2025)	0	3	3	1	3

Table 108: MRQ3 subsystem-coverage scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.									
Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Kannari et al. (2025)	3	1	1	1	1	1	2	1	2

Table 109: MRQ3 coordination-level and external-interface coverage for Kannari et al. (2025), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Kannari et al. (2025)	1	–	–

Table 110: MRQ3 gap-signal scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.						
Survey	HVAC	DHW/water	IAQ/ventilation/humidity	Lighting/plug loads	EV/V2H/V2G	Whole-building/BMS holistic
Kannari et al. (2025)	0	1	3	1	1	3

Table 111: MRQ3 summary table for Kannari et al. (2025), after recoding.			
Survey	Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Kannari et al. (2025)	Specialized real-building deployment and HVAC-centered	The review of real-building RL deployments is heavily concentrated on HVAC and related thermal-control applications, often in single-zone settings and with comfort represented mainly through indoor temperature. Broader services such as domestic hot water, lighting, plug loads, EV coupling, and richer IAQ-focused or personalized comfort control appear only in scattered cases.	For MRQ3, the paper is best interpreted as evidence that practical real-building RL deployments remain strongly HVAC-centric. The strongest under-reviewed directions are richer IEQ/IAQ-aware control and more holistic whole-building BMS perspectives beyond heating and cooling alone.

Table 112: One-line MRQ3 answer for Kannari et al. (2025), after recoding.	
survey_id	MRQ3 answer
KannariEtAl2025	Kannari et al. (2025) show that real-building RL deployments are still dominated by HVAC and thermal-control applications. IAQ-rich control, domestic hot water, lighting and plug loads, EV-coupled cases, and especially holistic whole-building BMS control remain sparse or explicitly identified as weakly represented directions.

17 Chatterjee and Khovalyg

Table 113: MRQ3 recoded audit table for Chatterjee & Khovalyg (2023). Scores use the shared 0–3 evidence-intensity scale.						
survey_id	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
ChatterjeeKhovalyg2023	0	3	3	0	2	The survey is specialized around reinforcement-learning-based HVAC controls for dynamic indoor thermal environments rather than broad all-building subsystem coverage. Broad scope receives score 0 because the paper is not a general subsystem survey. Main specialized scope receives score 3 because HVAC-based thermal-environment control is the declared and structuring scope. Secondary focus scope receives score 3 because dynamic indoor thermal environment control, including temperature step changes, drifts, thermal comfort, adaptation, and the energy–comfort–health framing, structures the paper. Multi-subsystem scope receives score 0 because broader non-HVAC services are outside the review’s scope. Deployment scope receives score 2 because real-building implementation, co-simulation, benchmarking, and open virtual testbed needs are discussed, but deployment is not the sole organizing scope.

Table 114: MRQ3 survey-scope scores for Chatterjee & Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Chatterjee & Khovalyg (2023)	0	3	3	0	2

Table 115: MRQ3 subsystem-coverage scores for Chatterjee & Khovalyg (2023), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Chatterjee & Khovalyg (2023)	3	1	1	1	1	1	1	1	1

Table 116: MRQ3 coordination-level and external-interface coverage for Chatterjee & Khovalyg (2023), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Chatterjee & Khovalyg (2023)	--	--	--

Table 117: MRQ3 gap-signal scores for Chatterjee & Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	HVAC	Dynamic temperature control/drifts	Humidity/air speed	IAQ	Broader non-HVAC services
Chatterjee & Khovalyg (2023)	0	3	1	1	0

Table 118: MRQ3 summary table for Chatterjee & Khovalyg (2023), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Chatterjee & Khovalyg (2023)	Specialized dynamic thermal-environment and HVAC-centered	The review is concentrated on HVAC-based thermal-environment control, especially the possibility of using RL to create dynamic temperature variations instead of static deadband control. Within that specialized scope, temperature is the dominant controlled variable, while humidity, air speed, and indoor air quality are acknowledged but much less central.	For MRQ3, the paper is best interpreted as evidence that a specialized strand of RL-for-buildings research is strongly HVAC- and temperature-centric. Even within this strand, RL for genuinely dynamic indoor thermal environments remains underdeveloped compared with conventional energy-saving and thermal-comfort optimization goals.

Table 119: One-line MRQ3 answer for Chatterjee & Khovalyg (2023), after recoding.

survey_id	MRQ3 answer
ChatterjeeKhovalyg2023	Chatterjee & Khovalyg (2023) show that this review strand is strongly centered on HVAC-driven thermal control, with dynamic temperature control as the structuring topic and temperature as the dominant controlled dimension. RL for genuinely dynamic indoor thermal environments remains comparatively underdeveloped, while broader non-thermal building services are outside the review’s scope.

Table 120: MRQ3 recoded audit table for Li et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
LiEtAl2025	0	3	2	1	2	Broad scope receives score 0 because the review is declared and organized around building-integrated photovoltaic optimization rather than building control generally, and its application chapter is partitioned by photovoltaic life-cycle stage covering design, generation forecasting, energy scheduling, storage scheduling, operation and maintenance, and system cost. Main specialized scope receives score 3 because that photovoltaic scope is the structuring principle of the entire review. Secondary focus scope receives score 2 because energy storage scheduling receives its own application subsection and storage recurs throughout the reward and state descriptions, without becoming a second organizing axis. Multi-subsystem scope receives score 1 because other building subsystems appear only as coordination targets named in passing: insufficient integration with heating, ventilation and air conditioning is listed as a limitation of current scheduling strategies, and one cited study is described as using specialized agents for heating, lighting and storage control. Deployment scope receives score 2 because real-world deployment receives a dedicated challenge subsection covering hardware limits, edge computation and building-management-system protocol incompatibility, but deployment is not itself a review axis.

Table 121: MRQ3 declared-scope scores for Li et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope
Li et al. (2025)	0	3	2	1	2

Table 122: MRQ3 subsystem-coverage scores for Li et al. (2025), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Li et al. (2025)	1	0	1	0	1	3	3	1	0

Table 123: MRQ3 coordination-level and external-interface coverage for Li et al. (2025), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Li et al. (2025)	1	2	1

Table 124: MRQ3 gap-signal scores for Li et al. (2025), recoded using the shared evidence-intensity scale and the normalised subsystem list.

Survey	HVAC	DHW/water	Lighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-building/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Windows
Li et al. (2025)	1	0	0	0	0	1	0	0	1	0	1	0

Table 125: MRQ3 summary table for Li et al. (2025), after recoding.

Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Generation and storage concentrated, with all other building subsystems at mention level or absent.	The review names insufficient integration between photovoltaic scheduling and building systems, specifically heating, ventilation and air conditioning together with storage, as a limitation of current rule-based strategies, and proposes coordination across distributed assets and scaling to district level as directions. It does not identify any subsystem as neglected by the literature it reviews.	For MRQ3, this survey contributes a coverage profile that is scope-determined almost in its entirety. Photovoltaic generation and battery storage reach structuring level because they are the declared subject; the zeros for domestic hot water, appliances and windows and the ones for lighting, heating and indoor air quality follow from that declaration rather than from any judgement about the field. The profile is therefore evidence about what a photovoltaic-scoped review covers, not about subsystem prevalence in the primary literature.

Table 126: One-line MRQ3 answer for Li et al. (2025), after recoding.

survey_id	MRQ3 answer
LiEtAl2025	Li et al. (2025) cover photovoltaic generation and battery storage at structuring level and grid interaction substantively, while heating, lighting, electric vehicles, whole-building management and indoor air quality appear only as passing mentions and domestic hot water, appliances and windows are absent; every one of those low scores follows from the declared photovoltaic scope rather than from a judgement about the literature, and the survey signals a gap only for integration between photovoltaic scheduling, heating systems and storage.

19 Togashi

Table 127: MRQ3 recoded audit table for Togashi (2025). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
Togashi2025	0	3	2	0	1	Broad scope receives score 0 because the review is declared and organized around reward function design for heating, ventilation and air conditioning control, and its screening criteria exclude studies outside that setting. Main specialized scope receives score 3 because that scope structures every analytical chapter, from the standardization of reward notation through the classification of comfort indicators to the proposed reward structure. Secondary focus scope receives score 2 because comfort assessment forms a distinct analytical axis within the review, with its own subsection quantifying which indicators the corpus uses and its own treatment of comfort limits, deadbands and acceptable ranges, without becoming a second declared scope. Multi-subsystem scope receives score 0 because no building sub-system other than heating and ventilation is covered at all: domestic hot water, lighting, appliances, electric vehicles and whole-building management do not occur anywhere in the survey, and storage and photovoltaic generation appear only as symbols in the notation table. Deployment scope receives score 1 because real buildings are invoked once, as the contrast case explaining why simulation studies do not surface equipment degradation from frequent cycling.

Table 128: MRQ3 declared-scope scores for Togashi (2025), recoded using the shared evidence-intensity scale.

Survey	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope
Togashi (2025)	0	3	2	0	1

Table 129: MRQ3 subsystem-coverage scores for Togashi (2025), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Togashi (2025)	3	0	0	0	0	1	1	2	2

Table 130: MRQ3 coordination-level and external-interface coverage for Togashi (2025), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Togashi (2025)	0	1	0

Table 131: MRQ3 gap-signal scores for Togashi (2025), recoded using the shared evidence-intensity scale and the normalised subsystem list.

Survey	HVAC	DHW/water	Lighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-building/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Windows
Togashi (2025)	0	0	0	0	0	0	0	0	0	2	0	0

Table 132: MRQ3 summary table for Togashi (2025), after recoding.

Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Single-subsystem, with indoor air quality and window operation as the only secondary presences.	Within comfort assessment rather than across subsystems. The review reports that dry-bulb temperature accounts for fifty-seven of the recorded indicator uses while carbon dioxide accounts for eight and particulate matter for two, and that sixty-eight of the seventy-nine studies rely on a single indicator, so air quality enters reward functions rarely and seldom alongside thermal measures.	For MRQ3, this survey has the narrowest coverage profile in the corpus, with five of the twelve subsystems at zero. Every one of those zeros follows from the declared scope, and none can be read as evidence about the field. The informative content lies elsewhere: within its single subsystem, the review measures how narrowly comfort itself is represented, and its treatment of window operation shows a subsystem entering the analysis as a formulation error rather than as coverage, since the argument is that window state belongs in the observation rather than in the reward.

Table 133: One-line MRQ3 answer for Togashi (2025), after recoding.

survey_id	MRQ3 answer
Togashi2025	Togashi (2025) covers heating and ventilation at structuring level and nothing else substantively, with indoor air quality and window operation appearing at discussion level and five of the twelve subsystems absent outright; the coverage profile is entirely scope-determined, and the survey’s only imbalance finding is internal to comfort measurement, where dry-bulb temperature dominates and carbon dioxide and particulate matter appear in a small minority of reward functions.

20 Lei et al.

Table 134: MRQ3 recoded audit table for Lei et al. (2026). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
LeiEtAl2026	0	3	3	3	3	Broad scope receives score 0 because the review is declared and organized around resilience of building multi-energy systems under disruption rather than around building control generally, and its application chapter is partitioned by resilience phase covering anticipation, maintenance of critical service during shocks, and restoration. Main specialized scope receives score 3 because that resilience framing structures the whole review, from the resilience metrics chapter to the paradigm-to-task mapping and the challenge and future chapters. Secondary focus scope receives score 3 because multi-energy carrier coupling forms a second organizing axis in its own right, with the interdependence of electricity, heat, cooling, gas and hydrogen driving the state and action formulation, the dimensionality challenge, and the scalable-learning future direction. Multi-subsystem scope receives score 3 because breadth across subsystems is constitutive of the review rather than incidental: resilience is defined as the capacity of an integrated building energy network spanning electricity, heating, cooling, ventilation and domestic hot water subsystems, and the storage state covers electrical, thermal and hydrogen carriers simultaneously. Deployment scope receives score 3 because real-world deployment receives its own application subsection with named installations and counted field validations, and a further future-direction subsection sets out three deployment pathways.

Table 135: MRQ3 declared-scope scores for Lei et al. (2026), recoded using the shared evidence-intensity scale.

Survey	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope
Lei et al. (2026)	0	3	3	3	3

Table 136: MRQ3 subsystem-coverage scores for Lei et al. (2026), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Lei et al. (2026)	2	1	0	0	0	3	2	1	0

Table 137: MRQ3 coordination-level and external-interface coverage for Lei et al. (2026), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Lei et al. (2026)	1	3	2

Table 138: MRQ3 gap-signal scores for Lei et al. (2026), recoded using the shared evidence-intensity scale and the normalised subsystem list.

Survey	HVAC	DHW/water	Lighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-building/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Windows
Lei et al. (2026)	0	0	0	0	0	2	0	0	1	0	2	0

Table 139: MRQ3 summary table for Lei et al. (2026), after recoding.

Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Energy-carrier centred, with storage and grid interaction at structuring level and end-use subsystems present but subordinate.	Across energy carriers rather than across building subsystems. The review reports that existing safe reinforcement learning work remains concentrated on single-carrier or electricity-heat systems, with limited systematic investigation of tightly coupled electricity, heat, gas and hydrogen systems under extreme events, and separately identifies cross-building generalization as underdeveloped.	For MRQ3, this survey occupies a position the corpus has not previously contained. It is not broad-scope, since resilience is a declared and structuring specialization, yet it covers more subsystems substantively than several broad-scope surveys because multi-carrier integration is constitutive of its subject. Its zeros for lighting, appliances, electric vehicles and windows are scope-determined, while its high storage and grid scores reflect that a building here is treated as an energy hub rather than as an occupied indoor environment. Any scope-stratified analysis has to place this survey deliberately rather than by default.

Table 140: One-line MRQ3 answer for Lei et al. (2026), after recoding.

survey_id	MRQ3 answer
LeiEtAl2026	Lei et al. (2026) cover storage and grid interaction at structuring level, heating, photovoltaic generation and building clusters substantively, and hot water, whole-building management and ventilation only in passing, with lighting, appliances, electric vehicles and windows absent; the profile follows from treating the building as a multi-carrier energy hub, and the imbalance the survey itself identifies runs across energy carriers, noting that safety-oriented work remains confined to single-carrier or electricity-heat systems.

21 Mazhar et al.

Table 141: MRQ3 recoded audit table for Mazhar et al. (2026). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
MazharEtAl2026	3	3	2	3	3	Broad scope receives score 3 because the review addresses smart building energy management across subsystems rather than restricting itself to one, treating heating, lighting, storage, generation and grid interaction together throughout its comparative tables and its control discussion. Main specialized scope receives score 3 because renewable energy optimization coupled with internet-of-things sensing is the declared organizing subject, named in the title and carried through the framework, validation and deployment chapters. Secondary focus scope receives score 2 because the sensing and communication layer forms a substantial but subordinate axis, covering sensor types, edge processing, federated architectures and communication protocols without becoming a second declared scope. Multi-subsystem scope receives score 3 because breadth is structural: the algorithm comparison tables record applications across heating, lighting and storage simultaneously, and scaling from single-zone to multi-zone settings organizes several subsections. Deployment scope receives score 3 because a staged roadmap from pilot testing to infrastructure optimization is given its own subsection, supported by economic feasibility analysis and named installations.

Table 142: MRQ3 declared-scope scores for Mazhar et al. (2026), recoded using the shared evidence-intensity scale.

Survey	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope
Mazhar et al. (2026)	3	3	2	3	3

Table 143: MRQ3 subsystem-coverage scores for Mazhar et al. (2026), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Mazhar et al. (2026)	3	1	3	1	1	3	3	1	1

Table 144: MRQ3 coordination-level and external-interface coverage for Mazhar et al. (2026), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Mazhar et al. (2026)	2	3	3

Table 145: MRQ3 gap-signal scores for Mazhar et al. (2026), recoded using the shared evidence-intensity scale and the normalised subsystem list.

Survey	HVAC	DHW/water	Lighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-building/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Windows
Mazhar et al. (2026)	0	0	0	0	0	0	2	1	1	0	2	0

Table 146: MRQ3 summary table for Mazhar et al. (2026), after recoding.

Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Broad and evenly weighted across generation, storage, grid interaction, heating, lighting and spatial scale, with end-use detail thin.	Scaling rather than subsystem neglect. The stated research gaps name difficulty scaling across multi-zone environments, variability in renewable sources complicating resource allocation and load balancing, underexplored integration between sensing infrastructure and control models, and an absence of real-world benchmarks. No subsystem is identified as neglected by the literature.	For MRQ3, this survey supplies the broadest coverage profile among the newly added reviews and one of the broadest in the corpus, with six subsystems at structuring level. The thin scores are informative in a different way from the specialized reviews: domestic hot water, appliances, electric vehicles and windows are not excluded by scope, since the review addresses building energy management generally, but appear only inside descriptions of cited studies. These are therefore coverage-determined rather than scope-determined low scores, and are usable as evidence about what a broad-scope review chooses to develop.

Table 147: One-line MRQ3 answer for Mazhar et al. (2026), after recoding.

survey_id	MRQ3 answer
MazharEtA12026	Mazhar et al. (2026) cover heating, lighting, storage, generation, grid interaction and multi-zone scaling at structuring level and whole-building management substantively, while domestic hot water, appliances, electric vehicles, indoor air quality and windows appear only inside summaries of cited studies; because the review declares a broad building energy management scope, those thin scores reflect what it developed rather than what its scope excluded, and the gaps it signals concern scaling, renewable variability and benchmark availability rather than any neglected subsystem.

22 Mohsenpour and Xing

Table 148: MRQ3 recoded audit table for Mohsenpour and Xing (2026). Scores use the shared 0–3 evidence-intensity scale.

survey_id	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
MohsenpourXing2026	0	3	3	3	3	Broad scope receives score 0 because the review is declared and organized around occupant-centric control of the indoor environment rather than around building energy management generally, and the energy supply side is absent: storage, photovoltaic generation and grid interaction occur once each or not at all. Main specialized scope receives score 3 because occupant-centric control is the structuring principle, with the reviewed literature classified as system-centric or occupant-centric and the proportions reported. Secondary focus scope receives score 3 because the co-optimization of energy, thermal comfort and indoor air quality forms a second organizing axis, with the results chapters partitioned by which combination of the three a study addresses. Multi-subsystem scope receives score 3 because three subsystems are declared in scope and each is developed at structuring level, with heating, lighting and window operation given their own action types in the action taxonomy, although the breadth is confined to the indoor environment. Deployment scope receives score 3 because real-world application receives its own subsection with quantified field results, followed by a data-sources subsection cataloguing the instrumentation and an implementation strategy figure.

Table 149: MRQ3 declared-scope scores for Mohsenpour and Xing (2026), recoded using the shared evidence-intensity scale.

Survey	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope
Mohsenpour and Xing (2026)	0	3	3	3	3

Table 150: MRQ3 subsystem-coverage scores for Mohsenpour and Xing (2026), recoded using the shared evidence-intensity scale and the normalised nine-subsystem list shared by all twenty-two surveys.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances/plug loads	EV/V2H/V2G	Battery/ESS	PV/renewables	IAQ/ventilation	Windows/blinds
Mohsenpour and Xing (2026)	3	1	3	0	0	1	1	3	3

Table 151: MRQ3 coordination-level and external-interface coverage for Mohsenpour and Xing (2026), recoded using the shared evidence-intensity scale. These three dimensions are reported separately from the subsystem table because they are levels of coordination rather than building subsystems; -- marks a dimension absent from that survey’s coding frame and not re-coded.

Survey	Whole-building/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Mohsenpour and Xing (2026)	1	1	1

Table 152: MRQ3 gap-signal scores for Mohsenpour and Xing (2026), recoded using the shared evidence-intensity scale and the normalised subsystem list.

Survey	HVAC	DHW/water	Lighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-building/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Windows
Mohsenpour and Xing (2026)	0	0	0	0	0	0	0	0	0	3	0	0

Table 153: MRQ3 summary table for Mohsenpour and Xing (2026), after recoding.

Coverage pattern	Strongest imbalance identified	Main MRQ3 consequence
Indoor-environment centred, with heating, lighting, window operation and air quality all at structuring level and the entire energy supply side absent.	Indoor air quality relative to thermal comfort. The review argues that the literature concentrates on keeping thermal comfort indices within acceptable ranges while minimizing energy, and that optimizing these in isolation can compromise air quality with consequences for occupant health, illustrated by an agent keeping windows closed to reduce cooling load while pollutants accumulate.	For MRQ3, this survey is the corpus counterpart to the energy-side specialized reviews. It reaches structuring level on four subsystems, more than most surveys in the corpus, yet scores at or below mention level on storage, generation, grid interaction, appliances and electric vehicles, because its declared scope is the indoor environment rather than building energy management. It also supplies the corpus’s clearest subsystem-level gap signal, since air quality is identified as systematically traded away rather than merely under-covered.

Table 154: One-line MRQ3 answer for Mohsenpour and Xing (2026), after recoding.

survey_id	MRQ3 answer
MohsenpourXing2026	Mohsenpour and Xing (2026) cover heating, lighting, window operation and indoor air quality at structuring level while the energy supply side is absent, with storage, generation, grid interaction and hot water appearing once each as table entries and appliances and electric vehicles not at all; the profile follows from a declared indoor-environment scope, and the survey supplies a genuine subsystem gap signal by arguing that air quality is systematically compromised when thermal comfort and energy are optimized in isolation.

23 Revision log: corrected coverage cells

The twenty-nine cells below had no column in the per-survey subsystem-coverage tables of the original audit trail and were filled with zeros when those tables were projected onto a shared list. Each was re-coded from the source publication during revision, and each correction is upward. Quotations are reproduced verbatim from the source.

Table 155: Coverage cells corrected during revision, with the evidence supporting each new score.

Survey	Subsystem	Old	New	Evidence
Mason & Grijalva (2019)	Windows/blinds	0	1	“Chen et al. applied Q learning to control both the HVAC and window systems [12]” (Sec. 4.1) — body coverage of one study, undeveloped
Han et al. (2021)	Appliances	0	1	“other behaviors, such as those associated with the use of cooking appliances, lighting, washing machines...” (Sec. 4.2.1) — an “other applications” list
Han et al. (2021)	IAQ/ventilation	0	2	“Valladares et al. [46] believed that occupant has strong influence on CO ₂ level and included the number of occupants as one of their states” (Sec. 4.1.1); further paragraph in Sec. 4.2.3; own Table 1 cells (state “levels of CO ₂ ”, action “setting temperature and ventilation system”)
Sierla et al. (2022)	DHW/water	0	2	“domestic water heating for houses is considered, with a single RL agent in each house taking binary decisions for controlling the heating element of a hot water tank” (Sec. 4.2.1); recurs in 4.1.1, 4.1.2, 4.2.2, 4.3.1
Sierla et al. (2022)	Lighting	0	1	“in some cases, indoor air quality of lighting comfort is being optimized” (Sec. 1) — describing prior reviews
Sierla et al. (2022)	Battery/ESS	0	1	“other kinds of HVAC applications for systems such as batteries [42] ... are out of scope” (Sec. 2) — named once, to exclude
Sierla et al. (2022)	PV/renewables	0	2	“Heating is provided by a ground source heat pump and solar thermal collector. RL is used to determine the mass flow rate of the water circulation loops to the solar thermal collector” (Sec. 4.2.2). Note: solar <i>thermal</i> ; PV-electric alone would be 1
Sierla et al. (2022)	Windows/blinds	0	2	“a RL agent that decides to open or close a window is the simplest solution for achieving the best tradeoff between indoor air quality and thermal comfort [96]” (Sec. 4.1.1); awning control in 4.1.4; joint window+HVAC control in 4.3.2.4
Chatterjee & Khovalyg (2023)	DHW/water	0	1	Table 1 System-Type cells for prior reviews [26] and [37]; “balance comfort, energy use, and hygiene in hot water systems” (Sec. 1)
Chatterjee & Khovalyg (2023)	Lighting	0	1	“A dynamic indoor environment includes but is not limited to, changes in temperature, humidity, lighting, ventilation, and air quality” (Sec. 1)
Chatterjee & Khovalyg (2023)	Appliances	0	1	“deep policy gradient (DPG) methods are studied to control many responsive needs, such as air conditioners, electric vehicles, and dishwashers” (Sec. 4)
Chatterjee & Khovalyg (2023)	EV/V2H/V2G	0	1	Table 3 controlled-parameter cell “air conditioner on/off, electric vehicle on/off” (Mocanu et al. row)
Chatterjee & Khovalyg (2023)	Battery/ESS	0	1	Table 1 System-Type cells for prior reviews [26] and [35] — the only two “batter” occurrences
Chatterjee & Khovalyg (2023)	PV/renewables	0	1	“DRL algorithms can dynamically adjust temperature set points based on external factors (e.g., outdoor temperature, occupancy, solar radiation)” (Sec. 5) — solar as weather input; PV-electric alone would be 0
Chatterjee & Khovalyg (2023)	Windows/blinds	0	1	Table 1 System-Type cells for prior reviews [26] and [30]
Weinberg et al. (2023)	Lighting	0	1	“This includes both electrical appliances and lighting... In this review, we limit the definition of BES to encompass heating equipment, thermal storage, emission systems, and ventilation and air conditioning” (Sec. 2) — named while being scoped out
Weinberg et al. (2023)	Appliances	0	1	“Zhang, et al. perform a limited study of transferring policies between buildings with varying user preferences and appliance parameters” (Sec. 6.1)
Weinberg et al. (2023)	PV/renewables	0	2	“evaluated on a task of controlling the temperature setpoints of air-to-water heat pumps in two large residential building equipped with PV panels... coordination of the photovoltaic production” (Sec. 7.1); recurs 7.2, 9
Weinberg et al. (2023)	IAQ/ventilation	0	2	“we limit the definition of BES to encompass heating equipment, thermal storage, emission systems, and ventilation and air conditioning, as well as their auxiliary systems, as shown in Fig. 2” (Sec. 2); ventilation-side control in 5.1.2, 6.1, 7.1
Weinberg et al. (2023)	Windows/blinds	0	1	“Wang and Hong review the use of RL for applications in building control such as window opening, lighting, and HVAC” (Sec. 1.2) — describes another review’s scope
Latoń et al. (2024)	IAQ/ventilation	0	1	“devices... included in the technical infrastructure of the building (e.g., heating, ventilation, air conditioning)” (Sec. 4) — acronym expansion only
Pinthurat et al. (2024)	Lighting	0	1	“Smart home systems automate and manage household functionalities such as lighting, ES, heating, security, and other appliances” (Sec. 2.1)

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Survey	Subsystem	Old	New	Evidence
Pinthurat et al. (2024)	IAQ/ventilation	0	1	“adjusting lighting, HVAC, and more to align with user preferences” (Sec. 4.4) — unexpanded HVAC acronym only
Kannari et al. (2025)	Battery/ESS	0	1	Table 2 row for Svetozarevic et al. [33], “Electrical vehicle charging/discharging... EV SoC”; Table 1 OptiWatti Solar storage control
Kannari et al. (2025)	PV/renewables	0	2	Table 1 dedicated row: “An AI-based energy management module... which directs the property’s solar electricity production to its own use” (Sec. 4.1)
Kannari et al. (2025)	Windows/blinds	1	2	“An & Chen [69]... controlling simultaneously a window, an air cleaner and an air conditioner” (Sec. 4.2.1.2), with its own Table 2 row
Michailidis et al. (2025)	Appliances	0	2	“MARL DQN optimized residential appliance scheduling with PV... reducing costs by 30%, peak demand by 18.2%” (Table 6, Lu et al. [97]); further rows in Tables 8, 9; Sec. 5.9
Michailidis et al. (2025)	IAQ/ventilation	0	1	“heating, ventilation, and air conditioning (HVAC)” (Sec. 1.1) — acronym expansion only; no air-quality vocabulary anywhere
Michailidis et al. (2025)	Windows/blinds	0	1	“building-integrated photovoltaics (BIPVs), where solar panels are seamlessly embedded into façades, roofs, and windows” (Sec. 2.1) — mounting surface, not control